

Chapter 1

Evaluation and Results

Having described the proposed three-tier hybrid system in Chapter ??, this chapter evaluates it empirically. The aim is twofold: to quantify how well each individual tier performs, and to show that the smart hybrid fusion improves upon every single tier on the metric that matters most for intrusion detection. Section 1.1 defines the evaluation parameters, that is, the metrics used and the composition of the evaluation set. Section 1.2 details the simulation method: the dataset preparation, the baselines selected for comparison and the rationale behind them, the algorithm hyperparameters, and the experimental procedure. The results are then reported in three parts: the performance of the individual detection tiers (Section 1.3), the performance of the hybrid fusion strategies (Section 1.4), and an error analysis based on confusion matrices (Section 1.5). Each result is presented with the corresponding tables and figures, accompanied by a detailed discussion that compares the methods and explains the observed behaviour.

1.1 Evaluation Parameters

The system performs binary classification, in which a request flagged as an attack is treated as the *positive* class and a benign request as the *negative* class. Its performance is therefore measured through the confusion matrix and the metrics derived from it. The confusion matrix, whose layout is shown in Table 1.1, cross-tabulates the predicted class against the true class and yields four counts. The number of true positives (TP) is the number of attacks that are correctly detected; the number of true negatives (TN) is the number of benign requests that are correctly passed; the number of false positives (FP) is the number of benign requests that are wrongly flagged as attacks (the *false alarms*); and the number of false negatives (FN) is the number of attacks that are missed (the *missed detections*). The two diagonal cells of the matrix therefore count the correct decisions, while the two off-diagonal cells count the two kinds of error.

Table 1.1: Layout of the binary confusion matrix used in this chapter.

	Predicted: Attack	Predicted: Clean
Actual: Attack	TP (correct)	FN (missed attack)
Actual: Clean	FP (false alarm)	TN (correct)

From these four counts, four metrics are reported for every configuration. *Precision* is the fraction of the requests flagged as attacks that are genuinely attacks; it is high when

there are few false alarms, and is defined as

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (1.1)$$

Recall, also called the detection rate or the true-positive rate, is the fraction of the actual attacks that the system manages to detect; it is high when there are few missed detections, and is defined as

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (1.2)$$

For reference, *accuracy* is the fraction of all decisions that are correct,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (1.3)$$

Precision and recall capture complementary kinds of error and are usually in tension: a detector can trivially achieve perfect recall by flagging every request (at the cost of precision), or near-perfect precision by flagging only the most obvious attacks (at the cost of recall). To summarise both in a single number, the F-measure takes their weighted harmonic mean. The general F_β score, in which the parameter β sets how many times more important recall is than precision, is

$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}. \quad (1.4)$$

Setting $\beta = 1$ weights precision and recall equally and gives the familiar F1 score, the plain harmonic mean,

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (1.5)$$

Setting $\beta = 2$ weights recall twice as heavily as precision and gives the F2 score, which after substituting $\beta = 2$ into Equation 1.4 simplifies to

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}}. \quad (1.6)$$

In this thesis the F2 score of Equation 1.6 is treated as the primary metric of merit. The reason is rooted in the cost asymmetry of the intrusion detection task: a missed attack (a false negative) may lead to a data breach with severe consequences, whereas a false alarm (a false positive) merely costs an analyst a few minutes of investigation. Because the two errors are not equally damaging, a metric that weights recall above precision reflects the operational priority better than the balanced F1 score does. Accuracy, defined in Equation 1.3, is reported only for completeness and is not used as a headline metric, because on a near-balanced dataset it is less discriminative than the precision–recall family and can mask a poor recall.

All headline numbers in this chapter are computed on a single, fixed evaluation set whose composition is given in Table 1.2. This set is the held-out 30% split of the balanced corpus; after log parsing, 28,767 lines were successfully evaluated, of which 13,643 are attacks and 15,124 are clean, giving an attack ratio of 47.43%. Because the two classes are close to balanced, both precision and recall are meaningful and neither is trivially inflated by class skew.

Table 1.2: Composition of the primary evaluation set.

Quantity	Value
Evaluated log lines	28,767
Attack lines	13,643
Clean lines	15,124
Attack ratio	47.43%

1.2 Simulation Method

1.2.1 Dataset preparation

The corpus combines three sources of HTTP traffic, as introduced in Section ??: the CSIC 2010 dataset, real attack payloads from the PayloadsAllTheThings repository, and automatically generated synthetic traffic. In total it contains 111,065 log lines, balanced to 50% attack and 50% clean so that the learning algorithms are not biased towards either class. The corpus is then partitioned as follows. The supervised tier (Tier 2) is trained on 66,984 labelled lines, of which 32,043 are attacks and 34,941 are clean. The unsupervised tier (Tier 3) is trained, following the one-class principle, on the 34,941 clean lines only. The remaining held-out split forms the evaluation set of Table 1.2.

Every line is processed by the same pipeline described in Chapter ??: it is parsed, its URL is decoded over up to three layers, and it is converted into the shared 22-feature vector. The ground-truth label of each line is obtained from a marker embedded in the synthetic traffic generator and the CSIC conversion, so that the evaluation labels are exact rather than estimated. Static-resource requests are whitelisted out of the learning tiers, as in production, so that the reported numbers reflect the system’s real operating behaviour.

1.2.2 Baselines and rationale

To place the proposed hybrid in context, it is compared against a set of baselines that span the design space. The baselines, and the reason each was chosen, are the following:

- (i) **Tier 1 regex rules**, as a pure misuse-detection baseline. It represents the classical signature approach and establishes the precision ceiling and the recall floor of the system.
- (ii) **Logistic Regression**, as a linear supervised baseline against the non-linear Random Forest. It was chosen because it is the model reported as best by Pham et al. [?] on CSIC 2010, which makes the comparison directly relevant to the related work.
- (iii) **Isolation Forest** and **One-Class SVM**, as alternative one-class anomaly detectors against the Local Outlier Factor. They were chosen so that the choice of the operational Tier 3 model rests on an explicit comparison rather than an assumption.
- (iv) **Simple Voting** (alert if any tier fires, that is, the logical OR of the three tiers), as a naive fusion baseline against the proposed smart consensus. It isolates the contribution of the consensus logic from the mere act of combining tiers.

This selection lets the evaluation answer three distinct questions: which supervised model is best, which unsupervised model is best, and whether the smart consensus genuinely improves on both the single tiers and a naive fusion.

1.2.3 Algorithm hyperparameters

The hyperparameters of every algorithm are fixed in advance and summarised in Table 1.3. The two supervised models use balanced class weights to counteract any residual class imbalance during fitting. The Random Forest and the Isolation Forest use 200 estimators. The Local Outlier Factor uses twenty neighbours in novelty mode so that it can score unseen requests. The One-Class SVM uses a radial-basis-function kernel. The models that are sensitive to feature scale (Logistic Regression and One-Class SVM) are preceded by a standardisation step, whereas the tree-based models are not. Every estimator is given a fixed random seed, which makes training and inference deterministic.

Table 1.3: Algorithm hyperparameters used in the experiments.

Model	Hyperparameters
Random Forest (Tier 2)	200 trees, unlimited depth, balanced class weights, fixed random seed
Logistic Regression (Tier 2)	lbfgs solver, up to 2000 iterations, balanced class weights, with feature standardisation
Local Outlier Factor (Tier 3)	20 neighbours, novelty mode, contamination = 0.50, with feature standardisation
Isolation Forest (Tier 3)	200 trees, contamination = 0.50, fixed random seed, with feature standardisation
One-Class SVM (Tier 3)	RBF kernel, $\nu = 0.60$, scale gamma, with feature standardisation

1.2.4 Experimental procedure

The experiment evaluates each configuration on the full evaluation set by a single exact per-sample pass: for every one of the 28,767 lines, the configuration’s decision is computed and tallied into the confusion matrix, from which the four metrics are derived. Because all estimators are seeded and the evaluation visits every sample exactly once (rather than sampling a subset), the procedure is deterministic; repeating it yields identical numbers, so no averaging over multiple runs is required and no error bars are reported. In total, six single-model configurations (regex, Random Forest, Logistic Regression, Isolation Forest, One-Class SVM, Local Outlier Factor) and the hybrid fusion strategies (simple voting and the smart consensus) are evaluated under identical conditions, which guarantees that the comparison is fair and that any difference is attributable to the method rather than to the data or the protocol.

1.3 Performance of the Individual Detection Tiers

The first experiment measures each detector in isolation. Table 1.4 reports the four metrics for the regex baseline (Tier 1), the two supervised models (Tier 2), and the three

one-class models (Tier 3). Figure 1.1 visualises the same data as four grouped bar charts, and Figure 1.2 ranks the models by F1 score.

Table 1.4: Performance of the individual detectors on the evaluation set (all values in %).

Detector	Type	Prec.	Recall	F1	F2
Regex rules (Tier 1)	Signature	99.61	24.07	38.77	28.37
Random Forest (Tier 2)	Supervised	94.80	94.76	94.78	94.77
Logistic Regression (Tier 2)	Supervised	85.82	74.90	79.99	76.86
Isolation Forest (Tier 3)	Unsupervised	63.02	93.14	75.17	85.01
One-Class SVM (Tier 3)	Unsupervised	62.27	91.38	74.07	83.57
Local Outlier Factor (Tier 3)	Unsupervised	88.28	95.13	91.57	93.67

Several observations follow from Table 1.4 and Figures 1.1 and 1.2. First, the regex baseline behaves exactly as the misuse-detection paradigm predicts: its precision is the highest of all detectors at 99.61%, meaning that almost everything it flags is truly an attack, but its recall is only 24.07%, meaning it misses roughly three out of four attacks. Consequently its F1 (38.77%) and F2 (28.37%) are the lowest in the table. This confirms that a signature-only system, while precise, is inadequate as a standalone detector, and motivates the learning tiers.

Second, among the supervised models the Random Forest is decisively better than Logistic Regression, achieving an F1 of 94.78% against 79.99%. The gap is largest in recall (94.76% versus 74.90%), which indicates that the linear decision boundary of Logistic Regression cannot capture the non-linear interactions among the 22 features as well as the ensemble of trees. This result is also consistent with, yet distinct from, Pham et al. [?], who found Logistic Regression best on a high-dimensional TF-IDF representation; on the compact 22-feature representation used here, the non-linear Random Forest is clearly superior, which justifies its choice as the primary Tier 2 model.

Third, among the one-class models the Local Outlier Factor is the clear winner, with an F1 of 91.57% compared with 75.17% for the Isolation Forest and 74.07% for the One-Class SVM. Interestingly, all three reach a high recall (above 91%), but the Isolation Forest and the One-Class SVM pay for it with a low precision (around 62–63%), whereas the Local Outlier Factor keeps precision high at 88.28%. The reason is that the density-based, locally adaptive criterion of the Local Outlier Factor draws a tighter boundary around the normal manifold than the global partitioning of the Isolation Forest or the single margin of the One-Class SVM, so it admits far fewer clean requests as false anomalies. This is precisely why the Local Outlier Factor was selected as the operational Tier 3 model.

1.4 Performance of the Hybrid Fusion Strategies

The second experiment evaluates whether fusing the tiers improves on the best single tier. Table 1.5 compares the three tiers against the two fusion strategies, and Figure 1.3 plots the same five configurations against a 90% reference line for each metric. Figure 1.4 shows the precision–recall trade-off as a scatter plot, and Figure 1.5 overlays the three strongest configurations on a radar chart.

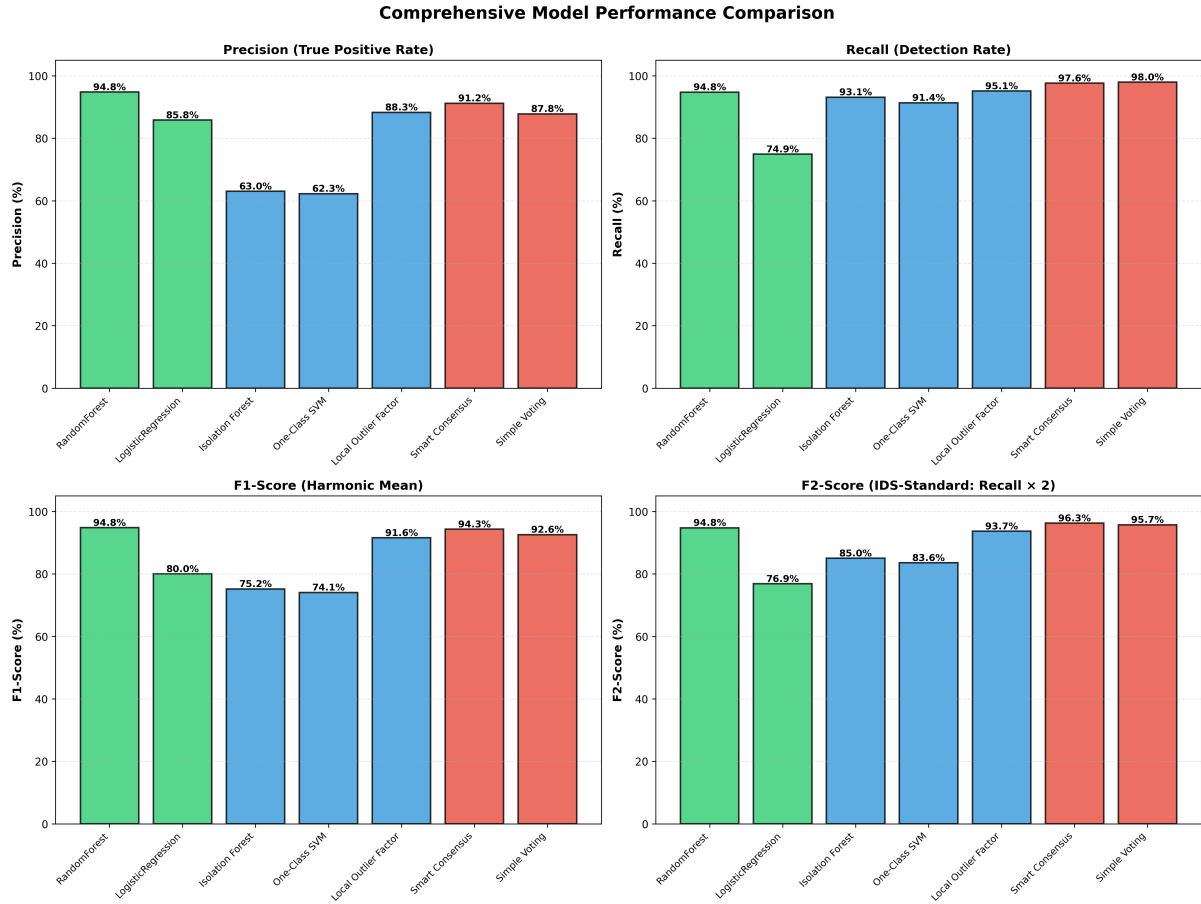


Figure 1.1: Precision, recall, F1, and F2 of every model, shown as four grouped bar charts. Green bars are supervised models, blue bars are unsupervised models, and red bars are the hybrid configurations (discussed in Section 1.4).

The central result of the thesis is visible in Table 1.5 and Figure 1.3. The proposed smart consensus attains the highest F2 score of the entire system at 96.26%, exceeding the best single tier (the Random Forest at 94.77%) and every other configuration. It does so by combining a recall of 97.62% with a precision of 91.19%: it catches about 98 of every 100 attacks while keeping roughly nine false alarms per hundred alerts. Compared with the Random Forest alone, the consensus trades a small amount of precision (from 94.80% to 91.19%) for a substantial gain in recall (from 94.76% to 97.62%); because the F2 metric rewards recall, this trade is favourable for an intrusion detection system, where missed attacks are the more dangerous error.

The comparison between the two fusion strategies explains why the consensus logic, and not mere combination, is responsible for the gain. Simple voting, which raises an alert whenever any tier fires, reaches the highest recall of all (97.96%) but its precision drops to 87.78%, because it inherits every false positive of the Local Outlier Factor. The smart consensus, by requiring that a Tier 3 anomaly be corroborated by at least a mild Random Forest suspicion before it is trusted, recovers precision to 91.19% while sacrificing almost no recall (only 0.34 percentage points). The net effect is a higher F1 (94.30% versus 92.59%) and a comparable F2, confirming that the supervised tier’s validation of the anomaly tier is the key mechanism. The precision–recall scatter in Figure 1.4 makes this trade-off geometric: the hybrid points are pushed towards the desirable top-right corner,

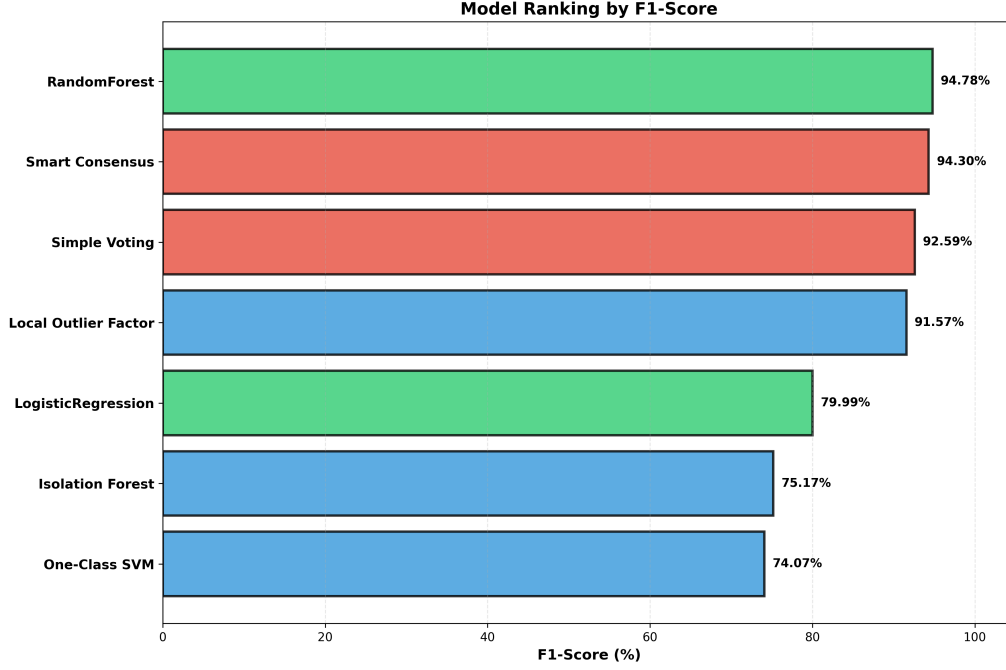


Figure 1.2: Ranking of the detectors by F1 score. The Random Forest leads, closely followed by the smart consensus and the Local Outlier Factor.

Table 1.5: Tiers versus hybrid fusion strategies on the evaluation set (all values in %). The best value in each column is in bold.

Configuration	Prec.	Recall	F1	F2
Tier 1: Regex rules	99.61	24.07	38.77	28.37
Tier 2: Random Forest	94.80	94.76	94.78	94.77
Tier 3: Local Outlier Factor	88.28	95.13	91.57	93.67
Simple Voting (T1 OR T2 OR T3)	87.78	97.96	92.59	95.74
Smart Consensus (proposed)	91.19	97.62	94.30	96.26

away from the low-precision cluster of the Isolation Forest and One-Class SVM.

It is worth noting how these figures relate to the related work surveyed in Chapter 2. Although a direct numeric comparison is imperfect because the evaluation protocols and feature sets differ, the proposed hybrid attains a recall (97.62%) that is higher than the attack-class recall reported by Pham et al. [?] for their best model, while operating directly on raw logs and, unlike the purely supervised studies, retaining an unsupervised tier capable of flagging novel attacks. The contribution of this thesis is therefore not a marginal accuracy gain on a fixed benchmark but a fusion design that is simultaneously recall-oriented, robust to unseen attacks, and explainable.

1.5 Error Analysis via Confusion Matrices

To understand *where* each configuration makes mistakes, this section examines the confusion matrices in Figure 1.6. Each matrix reports, for one configuration, the four counts TN , FP , FN , and TP ; the off-diagonal cells are the errors, with the bottom-left cell (FN) being the missed attacks and the top-right cell (FP) being the false alarms.

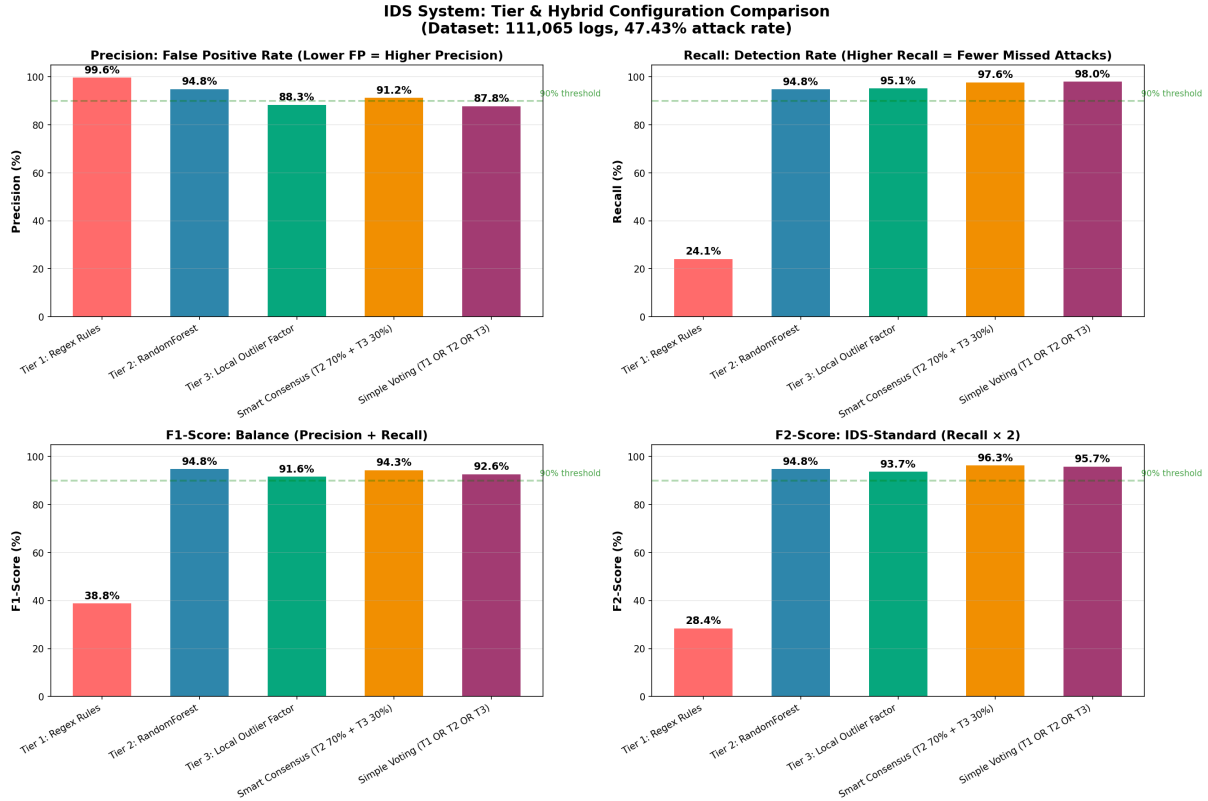


Figure 1.3: Tier and hybrid configuration comparison across the four metrics. The dashed line marks the 90% reference level. Both fusion strategies exceed every single tier on recall and F2.

The matrices tell a consistent error story, and because they are computed on the same evaluation set their cell counts can be read directly. The regex-only panel has an almost empty false-alarm cell (only 13 false positives) but a very large missed-attack cell (10,359 of the 13,643 attacks missed), the visual signature of high precision and low recall. The Isolation Forest and One-Class SVM panels show the opposite imbalance: their missed-attack cells are small (around 900–1,200) but their false-alarm cells are enormous (about 7,500 clean requests wrongly flagged), reflecting their low precision. The Random Forest panel is well balanced, with small and almost equal errors on both diagonals (715 missed attacks and 709 false alarms).

The two right-most panels make the effect of the fusion concrete. Simple voting achieves the fewest missed attacks of all (278) but pays with 1,860 false alarms, because it inherits every Local Outlier Factor false positive. The smart consensus keeps the missed-attack cell almost as low (325 missed attacks, against 665 for the Local Outlier Factor alone) while cutting the false alarms to 1,286, well below both simple voting and the Local Outlier Factor’s own 1,723. In other words, requiring the Random Forest to corroborate a Tier 3 anomaly removes roughly a quarter of the anomaly tier’s false positives at the cost of only a handful of additional missed attacks. This is exactly the behaviour an intrusion detection system should exhibit: it errs, when it must, on the side of raising an alert rather than letting an attack through, and the confusion matrices therefore corroborate the quantitative conclusion of Section 1.4 that the smart consensus offers the best recall-oriented balance.

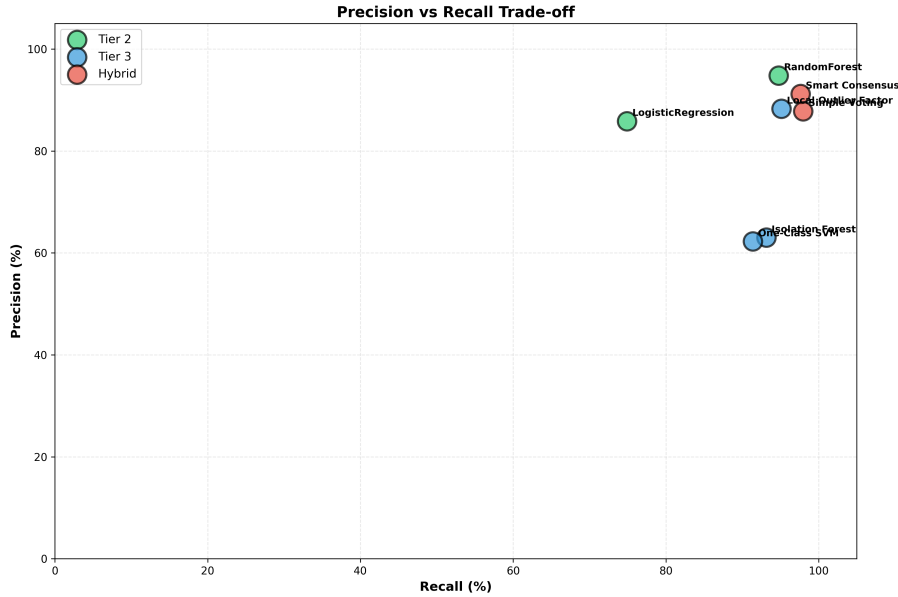


Figure 1.4: Precision–recall trade-off. The ideal corner is the top-right. The hybrid configurations sit in the high-recall, high-precision region, while the Isolation Forest and One-Class SVM occupy the high-recall but low-precision region.

1.6 Cross-dataset Evaluation and Dataset Selection

All results reported so far were obtained on the primary corpus, namely CSIC 2010 combined with the GitHub attack payloads and synthetic traffic. To check that the proposed methodology is not tied to a single benchmark, the entire three-tier pipeline was re-trained and re-evaluated, under an identical 70/30 protocol, on two additional public web-attack datasets: the HttpParams 2015 parameter-value dataset and the ECML/PKDD 2007 challenge dataset. Table 1.6 reports, for each dataset, the F1 score of the best supervised model (Random Forest) and the best unsupervised model (Local Outlier Factor), together with the F1 and F2 of the proposed smart consensus.

Table 1.6: The three-tier system trained and tested on each dataset under the same protocol (all values in %).

Dataset	Eval size	RF F1	LOF F1	Consensus F1	Consensus F2
CSIC 2010 + GitHub (primary)	28,767	94.78	91.57	94.30	96.26
HttpParams 2015	9,321	99.73	82.25	99.53	99.61
ECML/PKDD 2007	15,035	83.09	57.24	73.74	79.26

The three datasets behave very differently, and this is precisely what motivates the choice of CSIC 2010 as the primary corpus. On **HttpParams 2015** the system reaches almost perfect scores (Random Forest F1 of 99.73%), but this dataset consists of isolated parameter *values* built from textbook attack payloads, so the decision boundary between benign and malicious samples is unusually clean; in effect the dataset makes the system look “too easy” and does not reflect the structure of real HTTP traffic. On **ECML/PKDD 2007** the scores are the lowest (Random Forest F1 of 83.09%, consensus F2 of 79.26%) because its requests are heavily randomised: paths, parameters, and headers are synthetic, noisy strings, so even legitimate traffic looks irregular and the

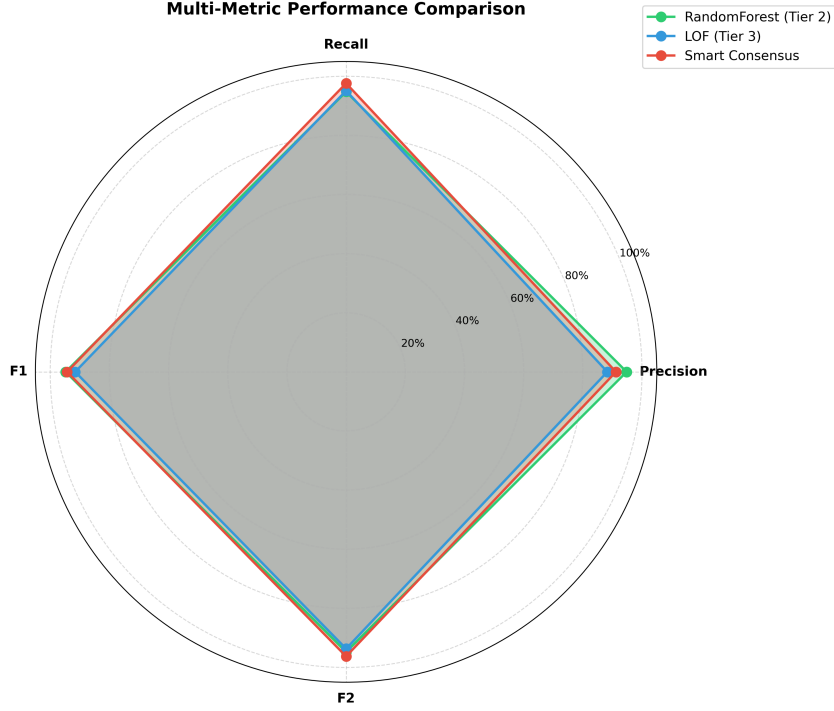


Figure 1.5: Multi-metric radar comparison of the three strongest configurations: the Random Forest, the Local Outlier Factor, and the smart consensus. The smart consensus dominates on the recall and F2 axes.

detectors struggle to learn a stable notion of “normal”. **CSIC 2010**, by contrast, was generated against a real e-commerce web application, so its requests have realistic endpoints, parameter names, and value distributions that closely match the traffic a deployed system would actually see; it yields strong and, above all, *representative* results (consensus F2 of 96.26%). For these reasons CSIC 2010 combined with the GitHub payloads was selected as the main dataset of the thesis: it offers the best balance between high, trustworthy accuracy and fidelity to real-world web traffic, whereas HttpParams is unrealistically easy and ECML is unrealistically noisy. Across all three datasets, however, the qualitative ranking of the tiers is consistent—the Random Forest is the strongest single model and the smart consensus delivers the highest F2—which confirms that the proposed methodology generalises and is not an artefact of one benchmark.

1.7 Chapter Summary

This chapter has evaluated the proposed system in full. Section 1.1 fixed the evaluation parameters: the precision, recall, F1, and F2 metrics, with F2 as the primary measure, on a balanced evaluation set of 28,767 lines. Section 1.2 described the simulation method, including the dataset preparation, the choice of the regex, Logistic Regression, Isolation Forest, One-Class SVM, and simple-voting baselines, the fixed hyperparameters, and the deterministic per-sample evaluation procedure. The results were then reported in three parts. The individual-tier experiment (Section 1.3) established the Random Forest as the best supervised model and the Local Outlier Factor as the best unsupervised model, and confirmed the high-precision, low-recall nature of the regex baseline. The hybrid experiment (Section 1.4) showed that the proposed smart consensus achieves the

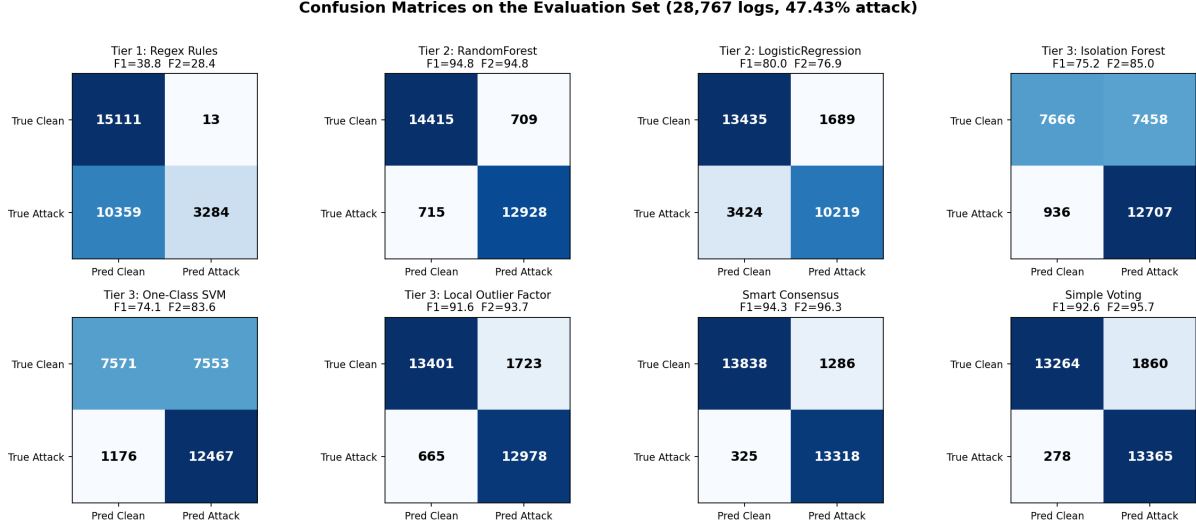


Figure 1.6: Confusion matrices of the eight configurations on the same 28,767-line evaluation set. For each panel the bottom-left cell counts missed attacks (FN) and the top-right cell counts false alarms (FP); the two diagonal cells are correct decisions. The per-panel F1 and F2 values match Tables 1.4 and 1.5.

system’s best F2 score of 96.26%, improving on every single tier and on naive voting by validating anomaly detections with the supervised model. Finally, the error analysis (Section 1.5) confirmed, through the confusion matrices, that the smart consensus minimises missed attacks while keeping false alarms under control. Finally, the cross-dataset study (Section 1.6) re-trained the pipeline on HttpParams 2015 and ECML/PKDD 2007 and justified the choice of CSIC 2010 as the primary corpus, showing that the tier ranking is consistent across datasets while CSIC offers the most realistic and trustworthy results. Together, these results demonstrate that the three-tier hybrid design meets its objective of recall-oriented, robust, and explainable web intrusion detection. The next chapter draws conclusions and outlines directions for future work.